

Routing tasks

The following figure shows the topology of a local network connected to the Internet. The local network consists of several network segments marked as N0, N1, ... (in red). Each router is marked R1, R2, ... and its network interfaces are marked a, b, c, d. Each router has a routing table, which always contains pairs - the destination network and the gateway. The destination network corresponds to the segment names (N1, N2, ...) or the DG value, which means that the destination is the Default Gateway. The second part represents the gateway (or its address) and consists of a combination of the router name and the network interface name separated by a dot, eg: R1.a or R3.d.

Your task is to fill in the routing tables so that all subnets can communicate with each other and the Internet. Assumptions:

- Assume that the R1 router has a working NAT and the Internet connection is guaranteed from this point of view.
- For the R1 router, you do not need to fill in the default gateway record.
- Do not fill directly connected segments into tables.
- Always specify the router address in the direction of the Internet as the default gateway.
- Save work and do not fill in routing tables with paths to segments that are already reachable through the default gateway (the solution may not be optimal, but it must be functional)
- Fill in the routing tables by entering the name of the router and then a pair of records (records do not have to be sorted), eg:

R9:

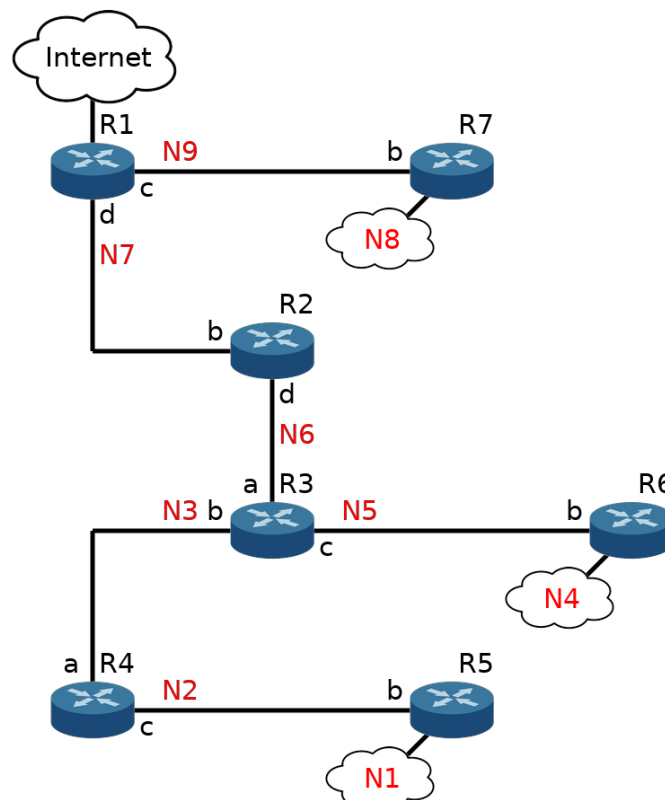
N1 -

DG - R8.d

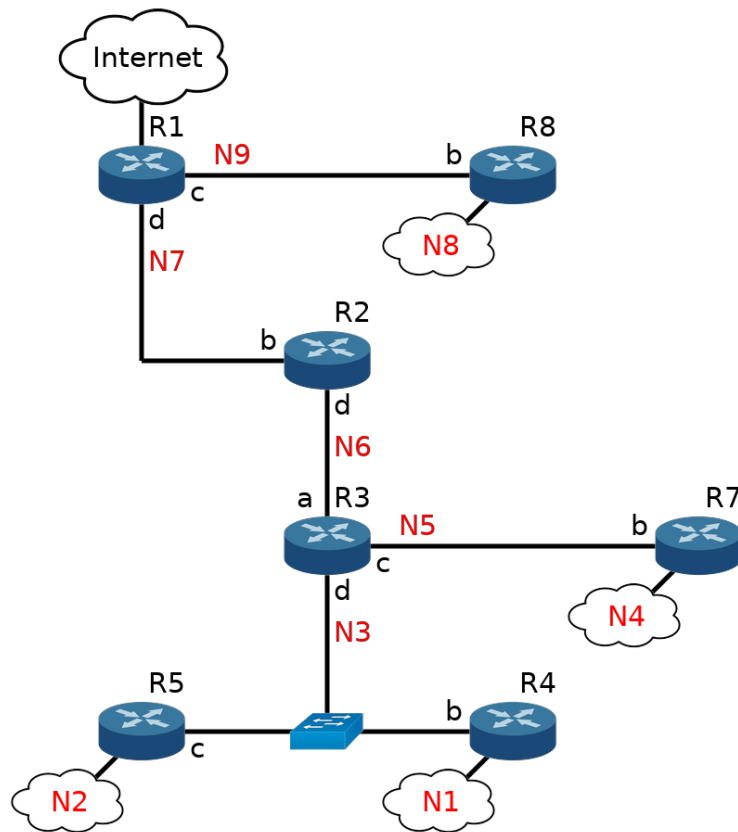
N2 - R10.a

...

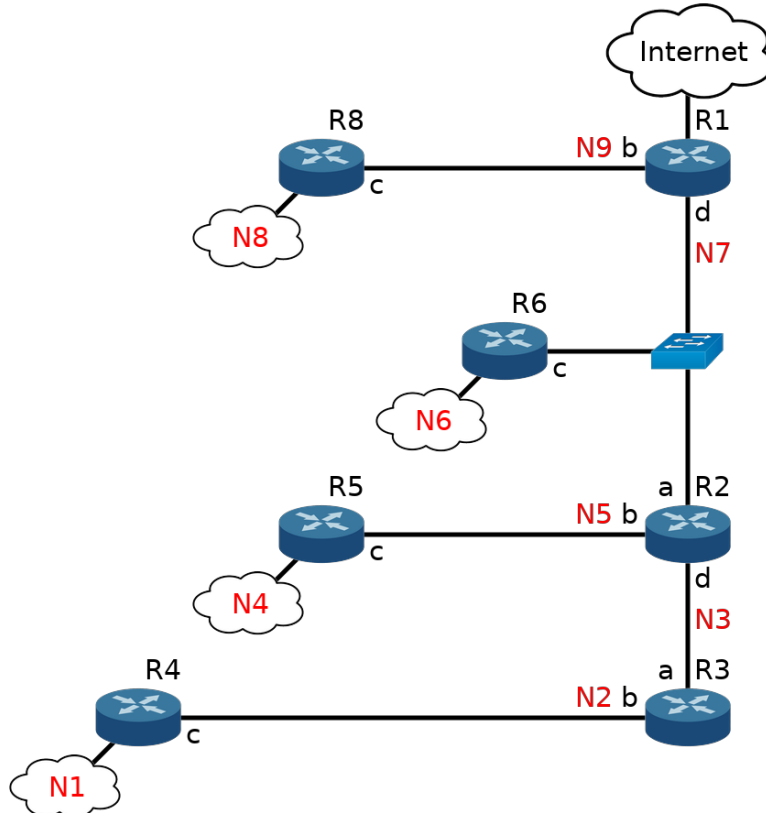
Task1.



Task2.



Task 3.



Solution is on the next page.

Task1.

R1

N1 R2.b

N2 R2.b

N3 R2.b

N4 R2.b

N5 R2.b

N6 R2.b

N8 R7.b

R2

DG R1.d

N1 R3.a

N2 R3.a

N3 R3.a

N4 R3.a

N5 R3.a

R3

DG R2.d

N1 R4.a

N2 R4.a

N4 R6.b

R4

DG R3.b

N1 R5.b

R5

DG R4.c

R6

DG R3.c

R7

DG R1.c

Task2.

R1

N1 R2.b

N2 R2.b

N3 R2.b

N4 R2.b

N5 R2.b

N6 R2.b

N8 R8.b

R2

DG R1.d

N1 R3.a

N2 R3.a

N3 R3.a

N4 R3.a

N5 R3.a

R3

DG R2.d

N1 R4.b

N2 R5.c

N4 R7.b

R4

DG R3.d

R5

DG R3.d

R7

DG R3.c

R8

DG R1.c

Task3.

R1

N1 R2.a

N2 R2.a

N3 R2.a

N4 R2.a

N5 R2.a

N6 R6.c

N8 R8.c

R2

DG R1.d

N1 R3.a

N2 R3.a

N4 R5.c

R3

DG R2.d

N1 R4.c

R4

DG R3.b

R5

DG R2.b

R6

DG R1.d

R8

DG R1.b